



Quantum Programming and Software

From core programming languages to community resources and beyond

The advent of quantum computing has not only revolutionised our understanding of computational theory but has also introduced a new paradigm in programming. Quantum programming encompasses the set of tools, languages, and methodologies designed to develop algorithms for quantum computers, offering a bridge between abstract quantum concepts and practical computational applications.

Introduction to Quantum Programming

Quantum programming is an emergent discipline that marries the principles of quantum mechanics with computer science, enabling the realisation of quantum algorithms that solve complex problems more efficiently than their classical counterparts. This section introduces the essentials of quantum programming, highlighting its importance and the evolution of the